Lesson 4

Reinforcement Learning

Reward & Penalty — How Computers Learn by Doing

Grade 4 · Academic Year 2025–26

Learning Objectives

1	Explain what Reinforcement Learning (RL) is using everyday examples.
2	Define the key RL terms: agent, environment, action, reward, and penalty.
3	Compare how humans and computers learn through trial and error.
4	Distinguish between positive reinforcement and negative reinforcement.
5	Apply RL thinking to make smarter choices in daily life.

What Is Reinforcement Learning?

Reinforcement Learning (RL) is a way of learning by trying things out and receiving feedback — rewards for good actions and penalties for unhelpful ones. The learner (called an agent) explores its environment, takes actions, and adjusts its behaviour based on the feedback it receives.

■ ***Just like learning to ride a bike — you keep trying, fall down (penalty), balance better (reward), and eventually ride smoothly!***

In computer science, RL is used to train AI agents to play games, control robots, manage energy in buildings, and even drive cars — all by trying millions of actions and learning from the results.

Real-Life Examples You Already Know

RL in Everyday Life

Situation	Agent	Action	Reward	Penalty
Riding a bike	You	Pedal & balance	Smooth ride!	Falling down
Training a dog	Dog	Sit on command	Yummy treat	No treat

Video game	Player	Jump over obstacle	Points / level up	Lose a life
Homework	Student	Finish on time	Teacher's praise	Extra homework
Chess AI	Computer	Move a piece	+score for winning	–score for losing

Key Terms in Reinforcement Learning

The vocabulary of RL:

Agent	The learner — the person, animal, or computer program that takes actions and learns from feedback.
Environment	The world the agent lives and acts in — a game board, a road, a classroom, or a market.
Action	What the agent does: move left, jump, say hello, submit homework, make a trade.
Reward	Positive feedback received after a good action — points, praise, a treat, money earned.
Penalty	Negative feedback after a poor action — losing points, a time-out, starting over.
Policy	The strategy the agent learns over time: 'In this situation, do THIS action to get the best reward.'

■ ***YOU are the agent. Your school, home, and playground are your environment. Every choice you make is an action — and life gives you rewards and penalties!***

How Computers Learn Like Us

Comparing human and computer learning:

HUMANS	COMPUTERS (RL agents)
Learn by practice and feedback	Try thousands of actions per second
Remember past successes and failures	Store everything in digital memory
Use feelings to judge good from bad	Use a numerical reward score
Improve with experience over years	Improve by replaying millions of games

Can learn from just one or two examples	Often need many repetitions to learn well

A key advantage computers have is speed: an RL agent can play a video game 1,000 times in a single minute, learning from each attempt. This is how AlphaGo (by DeepMind) became better than any human at the ancient game of Go — it played millions of games against itself.

Positive & Negative Reinforcement

Both types of reinforcement encourage good behaviour — but in different ways. Understanding the difference helps you recognise RL patterns in your own life.

Positive vs Negative Reinforcement

Type	What Happens	Effect	Example
Positive Reinforcement	Add something pleasant after good behaviour	Makes you more likely to repeat the behaviour	Get a gold star for tidy work
Negative Reinforcement	Remove something unpleasant after good behaviour	Also makes you more likely to repeat the behaviour	Parents stop reminding you once homework is done
Punishment	Add something unpleasant after bad behaviour	Makes you less likely to repeat the behaviour	Extra chores for breaking a rule

Important: Negative reinforcement is NOT punishment. It means removing something annoying as a reward for good behaviour!

Trial and Error — The Heart of RL

Learning through trial and error is one of the oldest and most powerful methods of learning. Every mistake is a lesson, not a failure. The key is to reflect on what went wrong and try a different approach next time.

The RL learning loop:

- 1. OBSERVE — Look at your environment. What is the current situation?
- 2. ACT — Choose an action based on what you know so far.
- 3. RECEIVE FEEDBACK — Did you get a reward or a penalty?
- 4. UPDATE — Adjust your strategy: do more of what worked, less of what didn't.
- 5. REPEAT — Keep looping until you master the task.

✏■ Activity: Reward or Penalty?

For each action below, decide: Reward (R) or Penalty (P)?

(a) You finish your homework early. ___

(b) You shout loudly during silent reading. ___

(c) You help a classmate who is stuck. ___

(d) You forget to say thank you. ___

(e) You eat all your vegetables at dinner. ___

Making Smarter Choices

Apply RL thinking in daily life:

- Think Before You Act — pause and consider what reward each option leads to.
- Learn From Mistakes — ask: What can I do differently next time?
- Remember What Worked — if a strategy earned a reward before, try it again in similar situations.
- Ask for Feedback — teachers, parents, and coaches are your 'environment' giving you rewards and penalties.
- Practice Consistently — the more you try, the faster your inner RL agent improves!

Quick Check ✓

1. Explain Reinforcement Learning in your own words. Give ONE example from your daily life.

2. What is the difference between an 'agent' and an 'environment' in RL? Give an example of each.

3. How is Positive Reinforcement different from Negative Reinforcement? Give one example of each.

4. Why can computers learn some RL tasks faster than humans? What are humans still better at?

5. Describe the five steps of the RL learning loop in order.

Lesson Summary

★ RL = learning by doing: take an action → receive feedback → adjust your strategy.

★ Key terms: Agent (learner), Environment (world), Action, Reward, Penalty, Policy.

★ Positive reinforcement adds something nice; negative reinforcement removes something unpleasant.

★ Computers can try thousands of actions per second — humans bring wisdom and common sense.

★ Every mistake is data. Use it to update your strategy and keep improving!